



Patient Profile

Yash M. Patel, 21 Years, Male

Summary of Results

The liver panel demonstrates an **abnormal** pattern as identified by the predictive model. Key findings include a markedly elevated ALT, a low albumin concentration and A:G ratio, and a low alkaline phosphatase level, while AST and bilirubin values remain within normal limits.

Detailed Test Explanations

AST (SGOT) measures an enzyme present in liver, heart, and skeletal muscle. Values within the 15–40 U/L range are considered normal; Yash's level of 16 U/L is within limits.

ALT (SGPT) is a more liver-specific enzyme; normal limits are 10–49 U/L. A result of 100.5 U/L is markedly high and suggests hepatocellular injury or inflammation.

Alkaline Phosphatase (ALP) reflects bone and biliary tract activity; the normal range is 30–120 U/L. A level of 15.4 U/L is low, potentially indicating malnutrition, vitamin D deficiency, or a bone disorder.

Bilirubin Total (0.60 mg/dL) and its fractions (direct 0.10 mg/dL, indirect 0.10 mg/dL) fall within normal ranges, indicating no evidence of significant hemolysis or biliary obstruction.

Total Protein (6.39 g/dL) is within the 5.70–8.20 g/dL reference interval, suggesting adequate overall protein synthesis.

Albumin (2.00 g/dL) is below the 3.20–4.80 g/dL reference range, revealing hypoalbuminemia.

A:G Ratio (0.10) is far below the 0.90–2.00 range, reflecting a disproportionate reduction in albumin relative to globulins, which can be seen in chronic liver disease or severe malnutrition.

Overall Interpretation

The combination of an isolated ALT elevation, low albumin, and a severely reduced A:G ratio points to impaired hepatic synthetic function and possible ongoing hepatocellular damage. The low ALP adds complexity, possibly reflecting nutritional deficiencies or a bone-related issue. The predictive model's abnormal flag underscores the need for clinical correlation and further assessment.

Recommendations

- **Reevaluate liver health:** Obtain viral hepatitis serology, screen for alcohol use, and review medications or supplements that may affect liver enzymes.
- **Nutritional assessment:** Evaluate dietary protein intake and consider a dietitian referral to address hypoalbuminemia.
- **Follow-up laboratory testing:** Repeat liver function tests, complete blood count, iron studies, and vitamin levels (especially vitamin D and B12) in 4–6 weeks.
- **Imaging:** If clinically indicated, perform abdominal ultrasound to assess for structural liver disease.
- **Lifestyle:** Encourage abstinence from alcohol, avoidance of over-the-counter hepatotoxic drugs, and promote a balanced diet rich in lean protein and essential micronutrients.

Disclaimer

This report is informational and should not replace professional medical advice. Consult a qualified healthcare provider for diagnosis and treatment.